THADOMAL SHAHANI ENGINEERING COLLEGE, BANDRA (W)

Computer Engineering Department

Practical exam Experiment List for AI

Course Code: **CSL604** Year/Semester: TE / VI

Course Name: **Artificial Intelligence Lab** Academic Session: JAN-APRIL 2026

1)	Implement **DFS** search algorithm in Python/Java.
2)	Implement **DLS** search algorithm in Python/Java.
3)	Implement **DFID** search algorithm in Python/Java.
4)	Implement **Breadth First Search** algorithm in Python/Java.
5)	Implement **Greedy Best First Search** algorithm in Python/Java.
6)	Implement **A*** search algorithm in Python/Java.
7)	Implement **Genetics algorithm** in Python/Java. Consider the optimization function F(x)= ((a+2b+3c+4d)-30)
8)	Implement **Hill climbing algorithm** in Python/Java. Consider the following equation y= sin x (or) $-x^2$ (or) -5x2+3x+6
9)	Implement **crossover operation in Genetics algorithm** in Python/Java. Consider a population of 6 chromosomes and convert chromosomes into binary number and perform cross over operation , consider crossover rate as 25 %
10)	Implement **Mutation operation in Genetics algorithm** in Python/Java. Consider a population of 6 chromosomes and convert chromosomes into binary number and perform mutation operation , consider mutation rate as 10 %.
11)	Implementing Family tree using prolog.

Note:

1) Viva can be asked on entire syllabus Or technical paper that you have taken during practical's.
2) Prepare Block world problem, Wumpus world problem and First Order Logic, you can be given any sentence to be converted to FOL
3) For search algorithms draw a tree / graph on answer book and write output with open list and close list.
4) For Genetic algorithm take 6 chromosomes in population for performing all the operations or specified operation, in output u r expected to print initial population (6 chromosomes)and final population (6 chromosomes) and best chromosome from final population which is optimal solution for equation considered.
5) For 9th and 10th experiment perform one iteration.

Subject In charge
Dr. Shilpa Verma